Assignment

Q1. Variance and Bias (Diagram, overfit, underfit) - For best fit model should we have low bias or high variance, low bias or low variance, high bias or high variance, low bias or high variance.

INTRODUCTION

In machine learning, bias and variance are two fundamental sources of error in predictive models. Understanding their relationship is crucial for building models that generalize well to new, unseen data.

- **Bias**: Error introduced by approximating a real-world problem with a simplified model
- **Variance**: Error introduced by the model's sensitivity to fluctuations in the training data

BIAS-VARIANCE TRADEOFF

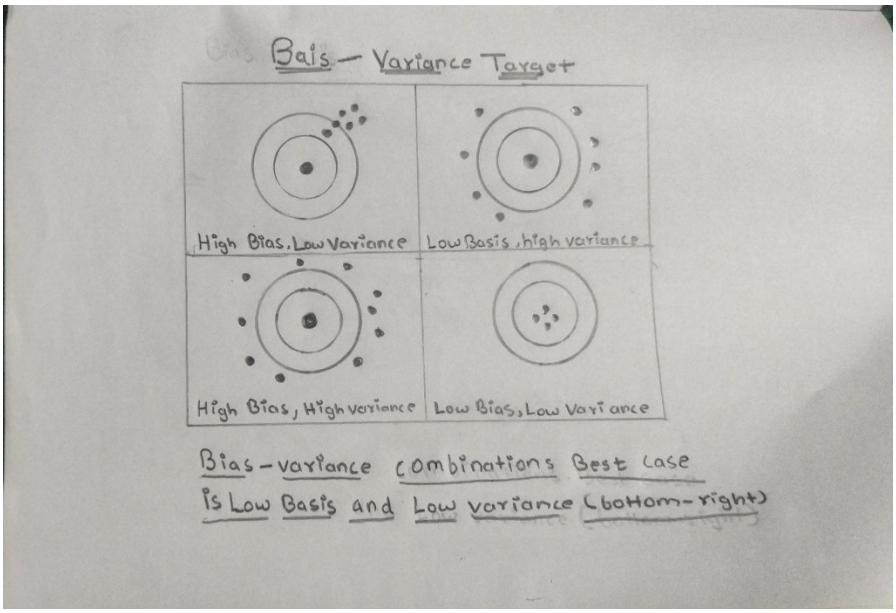
The bias-variance trade-off represents the challenge of simultaneously minimizing two sources of error that prevent supervised learning algorithms from generalizing beyond their training set.

BIAS-VARIANCE COMBINATIONS

The Four Possible Combinations

Combination	Bias	Variance	Model Behaviour	Performance
Combination 1	High	Low	Underfitted	Poor
Combination 2	Low	High	Overfitted	Poor

Combination 3	High	High	Poor fit	Very Poor
Combination 4	Low	Low	Best fit	**Optimal**



MODEL FITTING SCENARIOS

Three Types of Model Fitting

A. Underfitting (High Bias)

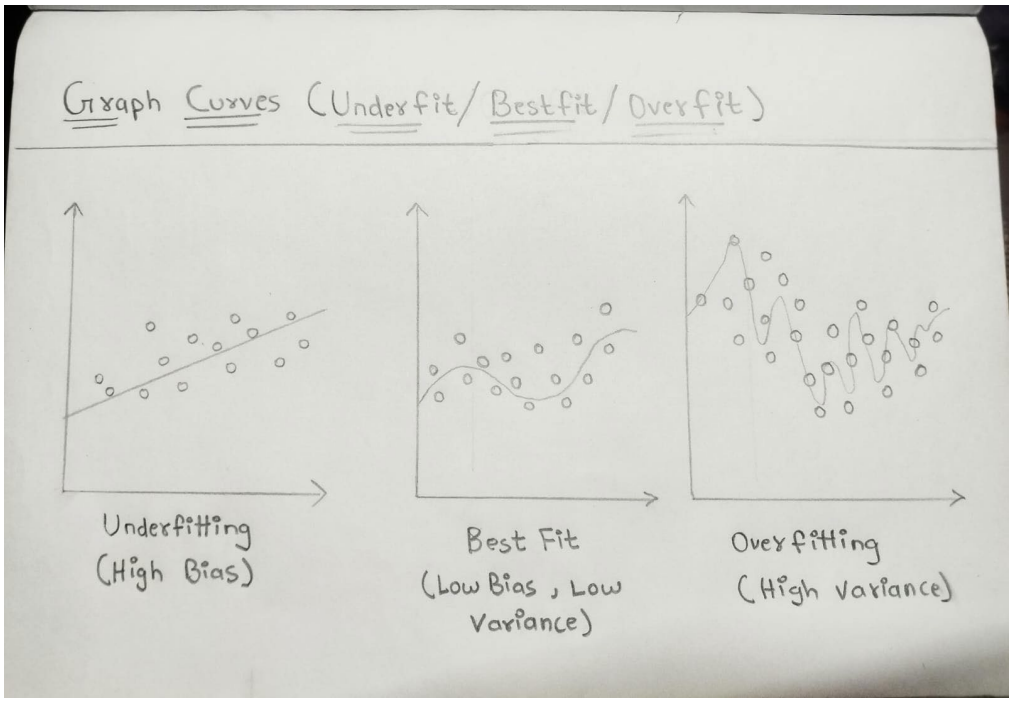
- Model is too simple
- Fails to capture underlying patterns in data
- Poor performance on both training and test data
- **Example**: Using linear regression for non-linear data

B. Best Fit (Low Bias, Low Variance)

- **Optimal model performance**
- Captures true underlying patterns
- Generalizes well to new data
- Perfect balance between bias and variance

C. Overfitting (High Variance)

- Model is too complex
- Learns noise in training data
- Excellent on training data, poor on test data
- **Example**: Very high-degree polynomial fitting



ANALYSIS OF BEST FIT MODEL

Answer to Assignment Question

Question: For best fit model should we have:

- Low bias or high variance?
- Low bias or low variance?
- High bias or high variance?
- Low bias or high variance?

Answer:

For a **best fit model**, we need:

LOW BIAS - The model must be complex enough to capture the underlying patterns in the data

LOW VARIANCE - The model should be stable and not change dramatically with small changes in training data

Therefore: The best fit model requires LOW BIAS and LOW VARIANCE

This combination (Low Bias, Low Variance) represents the ideal scenario where:

- The model accurately represents the true relationship (Low Bias)
- The model is consistent across different training samples (Low Variance)

COMPARISON TABLE

Model Performance Comparison

Model Type	Bias Level	Variance Level	Training Error	Test Error	Generalization
Underfitted	**High**	**Low**	**High**	**High**	**Poor**
Best Fit	**Low**	**Low**	**Low**	**Low**	**Excellent**
Overfitted	**Low**	**High**	**Very Low**	**High**	**Poor**
Poor Fit	**High**	**High**	**High**	**Very High**	**Very Poor**

CONCLUSION

The bias-variance trade-off is a fundamental concept in machine learning. The optimal model achieves a balance by maintaining **both low bias and low variance**. This ensures the model captures true patterns while remaining stable across different datasets, leading to excellent generalization performance on unseen data.